Tudy HERRIOU

Paris, France — +33(0)623459231 — tudy.herriou@polytechnique.edu — github.com/tudyh

Education

ENS Paris-Saclay **2025 – 2026**
Master Mathématiques, Vision, Apprentissage (MVA)
Relevant coursework: Convex Optimization, Reinforcement Learning, ML for Time Series, Probabilistic Graphical Models, Generative Modeling, LLM for code and proof, Training and deploying Large-Scale models.

Ecole Polytechnique (X) **2022 – 2026**
Master's degree in applied mathematics, graduation in June 2026. GPA 3.94/4.

Lycée Hoche **2019 – 2022**
Preparatory class for Grandes Ecoles, MPSI/MP*
Ranked 4th at the entrance examinations for Ecole Polytechnique, GPA 4/4.

Experience

Instadeep – DeepPCB Team **04/2026 – Present**
Research internship

- Contributed to leveraging Reinforcement Learning for electronic hardware design solving combinatorial problems.
- Designed a new pre-training strategy for the routing agent achieving 90% of baseline performance with 50% reduction of training time.
- Currently enhancing inference performance by architecting a novel value function network and generating custom synthetic datasets for targeted training.

Columbia University - New York Genome Center - Vickovic Lab **2025**
Research internship

- Developed an ML model using Graph Neural Networks to perform multimodal identification of the mouse brain using 2-photon and functional imaging under the supervision of Prof. Sanja Vickovic.
- Designed the model to accomplish Neural Subgraph Matching, created a pipeline generating synthetic data for pre-training and fine-tuned the Curriculum Learning approach initially used.
- Performed data cleaning and feature engineering, improving model accuracy on the real dataset from 0.73 to 0.89.

Stellantis **2024**
Research internship

- Developed a machine learning model for uncertainty quantification of acoustic vibrations in vehicles based on Random Matrix Theory.
- Implemented an optimization algorithm to estimate the dispersion parameters associated with the mass, stiffness, and damping matrices of the stochastic physical model.

Personal & Scientific Projects

Pseudoinverse-Guided Diffusion Models (ΠGDM) for Inverse Problems — *ENS Paris-Saclay* **2026**

- Implemented and evaluated ΠGDM, a zero-shot, problem-agnostic framework that estimates conditional scores from measurement models without requiring additional training.
- Conducted a comparative analysis—both theoretically and experimentally—against baseline methods (Chung et al., ICLR 2023) using DDPM architectures for image restoration tasks.
- Benchmarked model performance and numerical stability across linear and non-linear inverse problems, specifically targeting image inpainting and Gaussian deblurring under escalating measurement noise levels.

Diffusion Schrödinger Bridge (DSB) Analysis — *ENS Paris-Saclay* **2025**

- Explicited the closed-form solution of the discrete dynamic Schrödinger Bridge (SB) problem under the Gaussian case.
- Leveraged this mathematical framework to evaluate the performance of DSB models across various source and target distribution complexities (p_{data}, p_{prior}) and dimensions.
- Demonstrated theoretically and practically that the default setup of $L = 20$ iterations in the original benchmark leads to model under-training.

GPT-2 Language Model from Scratch — *Ecole Polytechnique* **2025**

- Rebuilt the GPT-2 Transformer architecture from scratch using PyTorch.
- Collected and cleaned a dataset of tweets related to football matches.
- Fine-tuned the model for sentiment classification (positive / negative / neutral).

Extra-Curricular Activities

X-Forum **2023 – 2025**
President of the Industry division

- The X-Forum is a meeting place for more than 150 companies and 2,000 students.
- Responsible for the prospection of Industry companies, managing a team of 12 persons.

Skills and Interests

Language abilities: French (native), English (fluent), Spanish (fluent)

Computer skills: PyTorch (Advanced), Jax (Intermediate), Pack Office (Intermediate)

Sports: Boxing (5 years at a competitive level)